New Grammer

Program : Program identifier ';' Block ';'

Block : {Var_List} {Function_definition }* Function_Body

Var_List : 'VAR' Data_ Definition { Data_ Definition }*

Data_ Definition : identifier_List ':' 'INTIGER' ';'

identifier_List: identifier { ',' identifier }*

Function_Definition : Function_Heading ';' block ';'

Function_Heading : 'FUNCTION' identifier { Parameter_list } ':' 'INTEGER'

Parameter_List : '(' identifier_List ':' 'INTEGER' ')'

Function_Body : 'BEGIN' Statement_Sequence 'END'

Statement_Sequence : Statement { ';' Statement }*

Statement : { identifier ':=' Expression }

| 'BEGIN' Statement_Sequence 'END'

| 'IF' Expression 'THEN' Statement { 'ESLSE' Statement }

| 'WHILE' Expression DO statement

| 'FOR' identifier ':=' Expreission ['TO' | 'DOWN TO'] Expression DO Statement

Expression : Simple_Expr | Simple_Expr Relop SSF

Relop : '=' | '<>' | '<' | '>' | '<=' | '>='

Simple_Expr : Oprand | Sign SSF2 | Simple_Expr Addop SSF2

Sign : '+' | '-'

Addop : '+' | '-'

Oprand : Factor | Oprand Multop Factor

Multop : '*' | '/' | 'DIV' | 'MOD'

Factor : identifier | Constant | '(' Expression ')' | identifier '(' Expression ')' | identifier '(' Expression SRF ')'

SSF : Simple_Expr

SSF2 : Operand

TF : ',' Expression TF | ',' Expression

SRF : TF

TF -> Tree Fasible

SRF -> Shift Reduce Fix

SSF -> Shift Shift Fix

SSF -> Shift Shift Fix 2